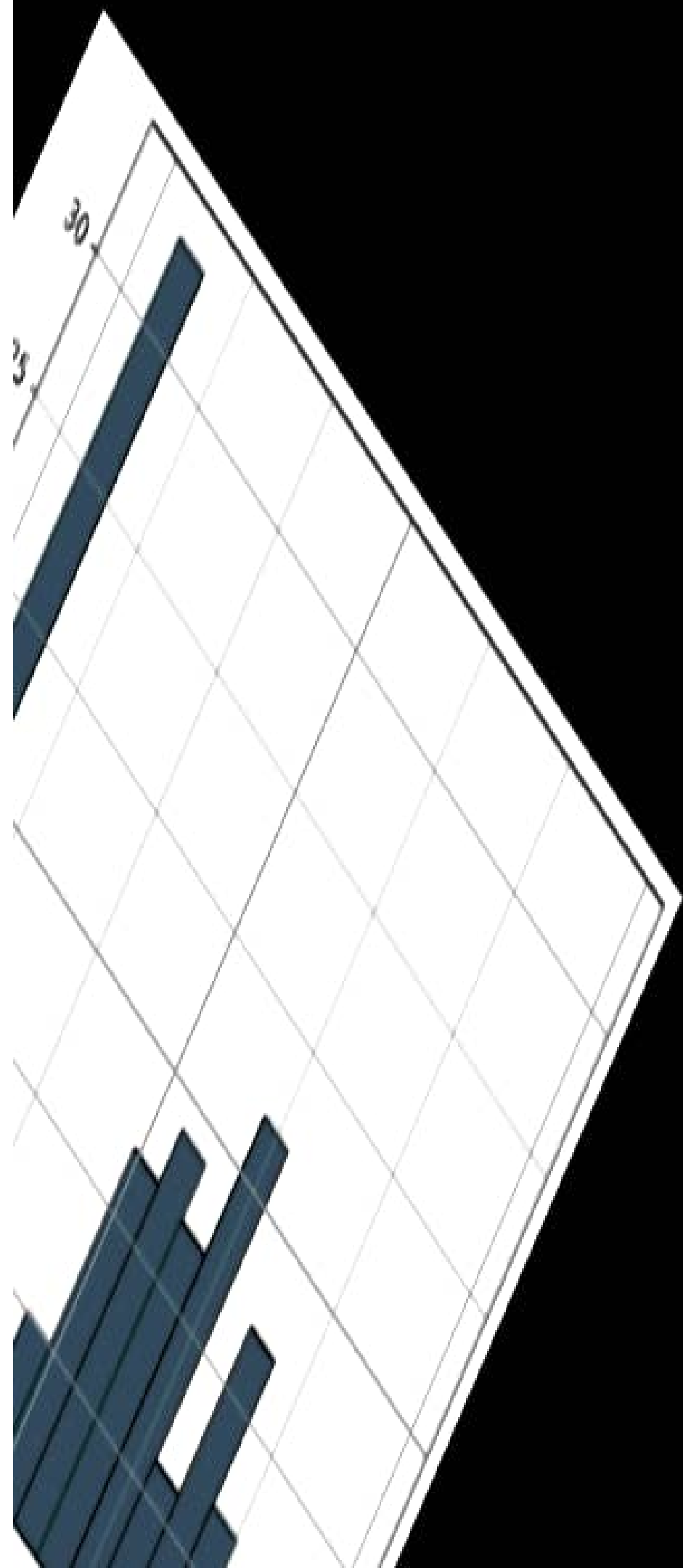
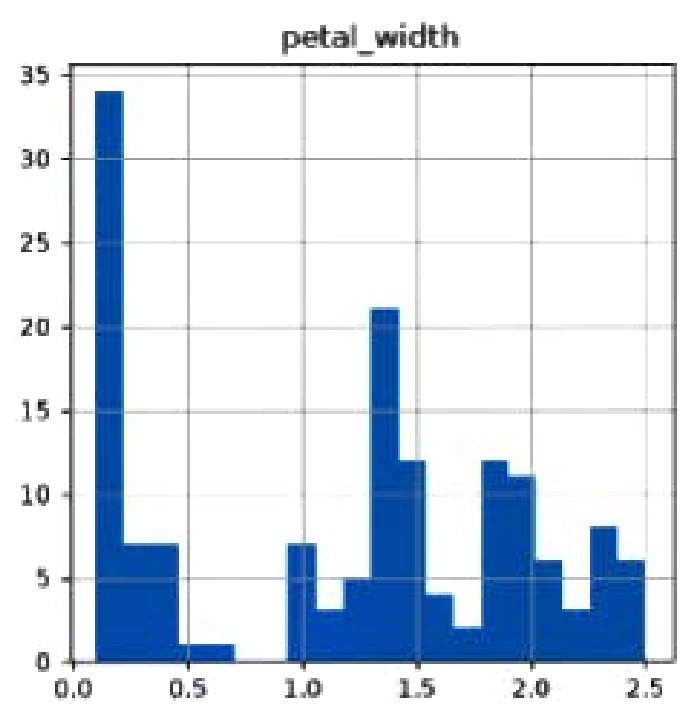
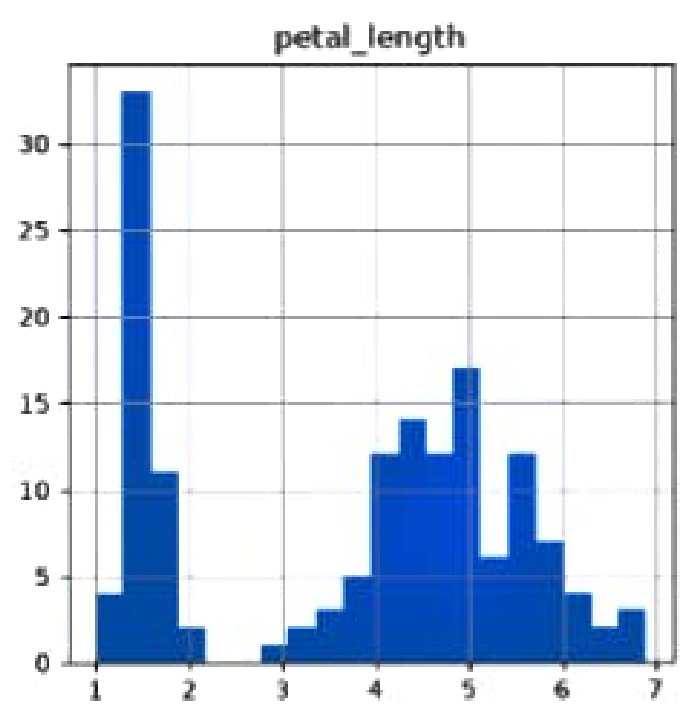
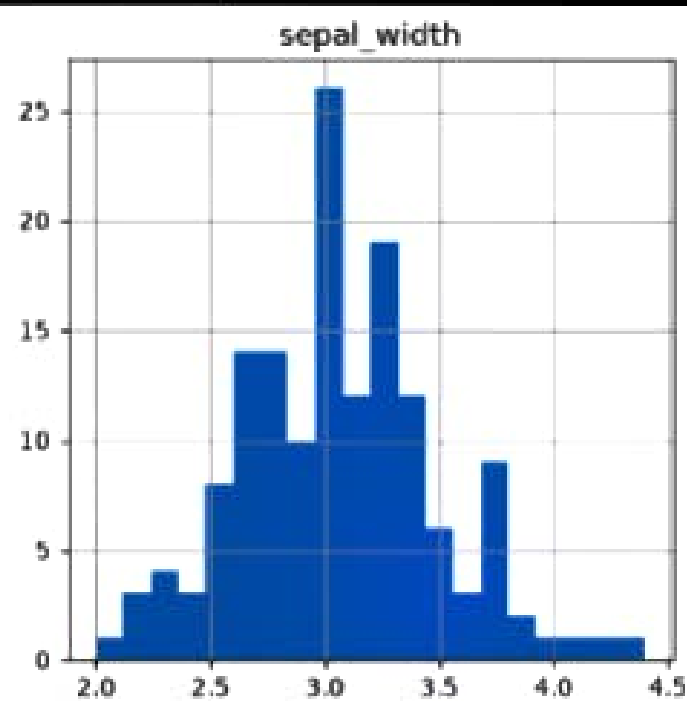
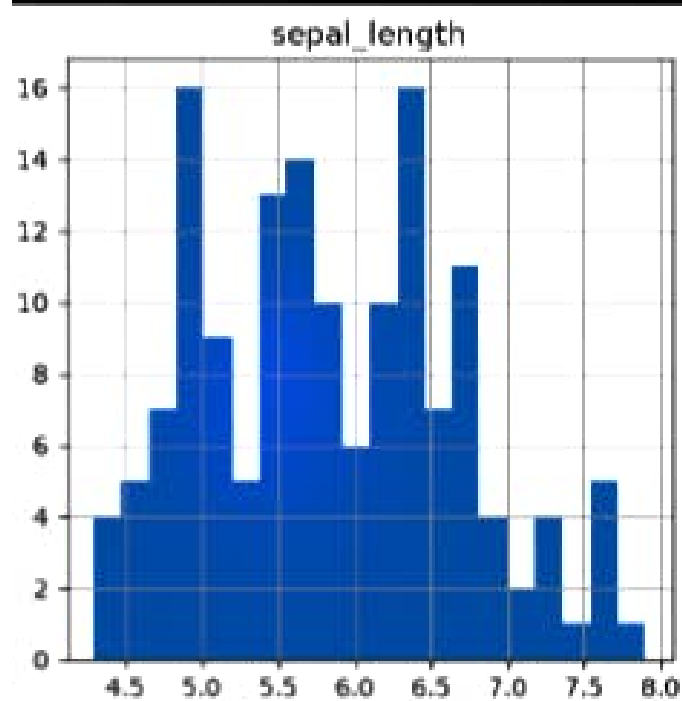
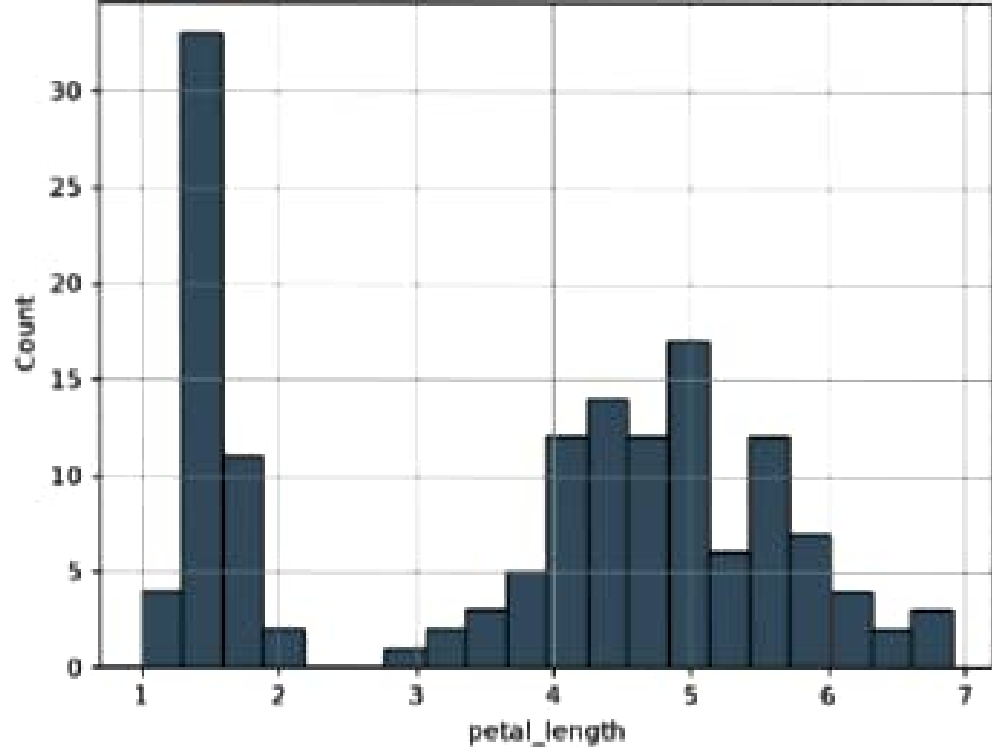
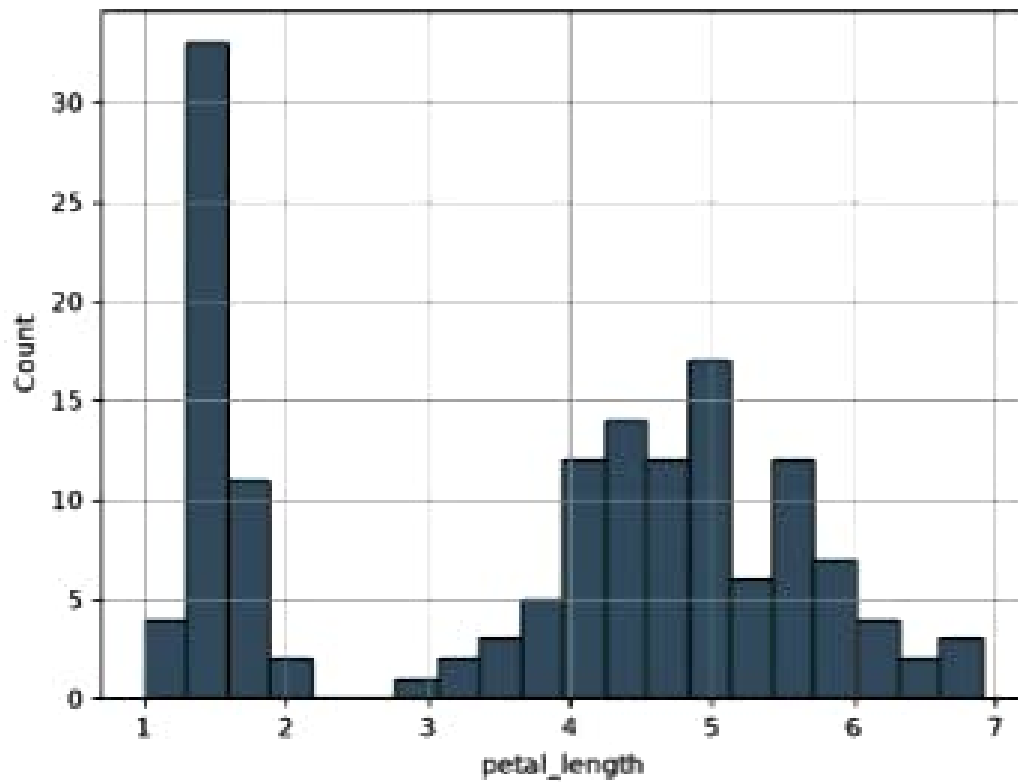
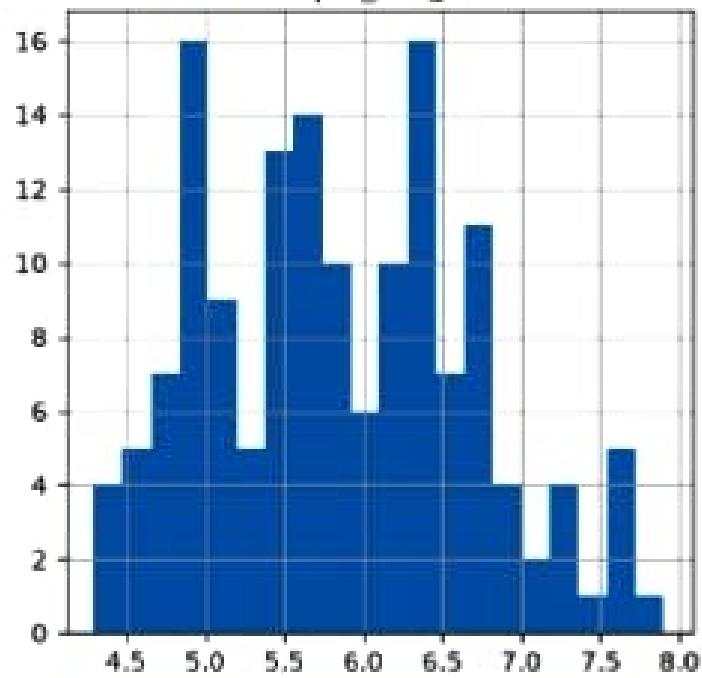




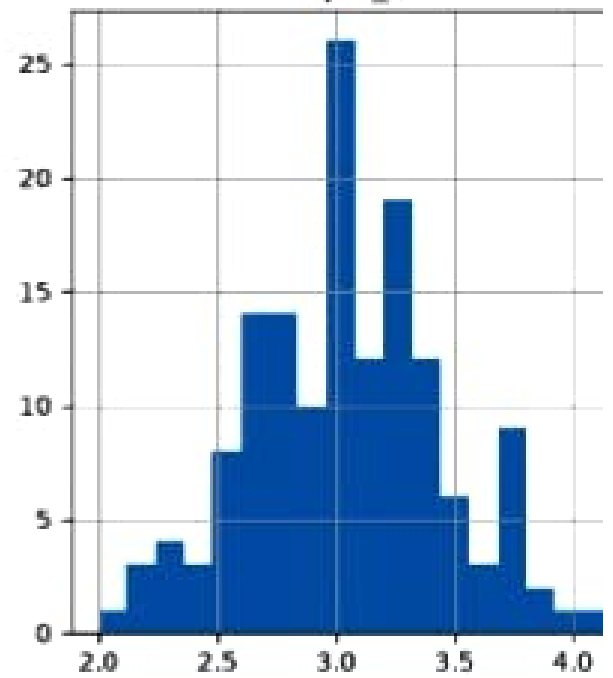




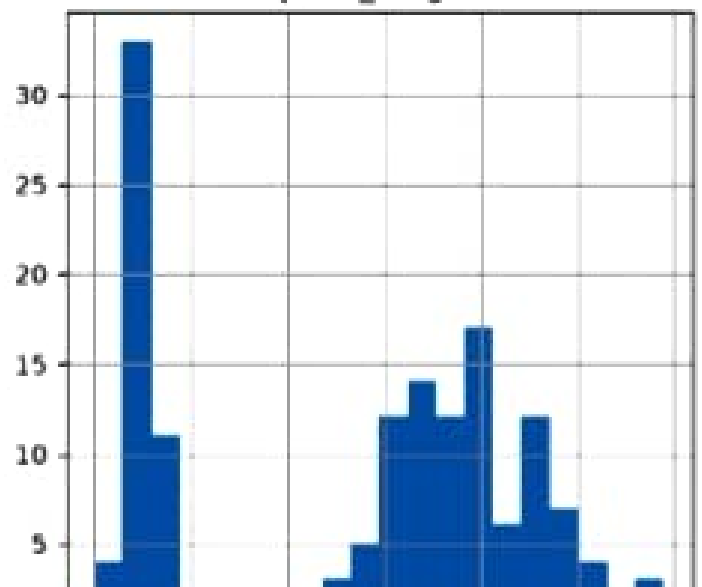
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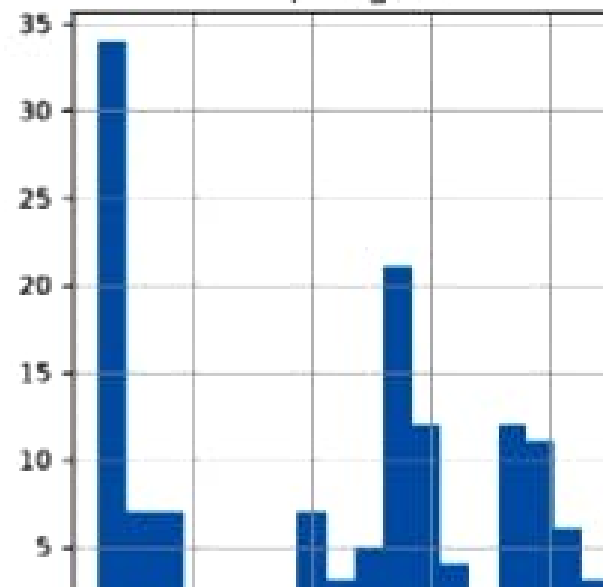
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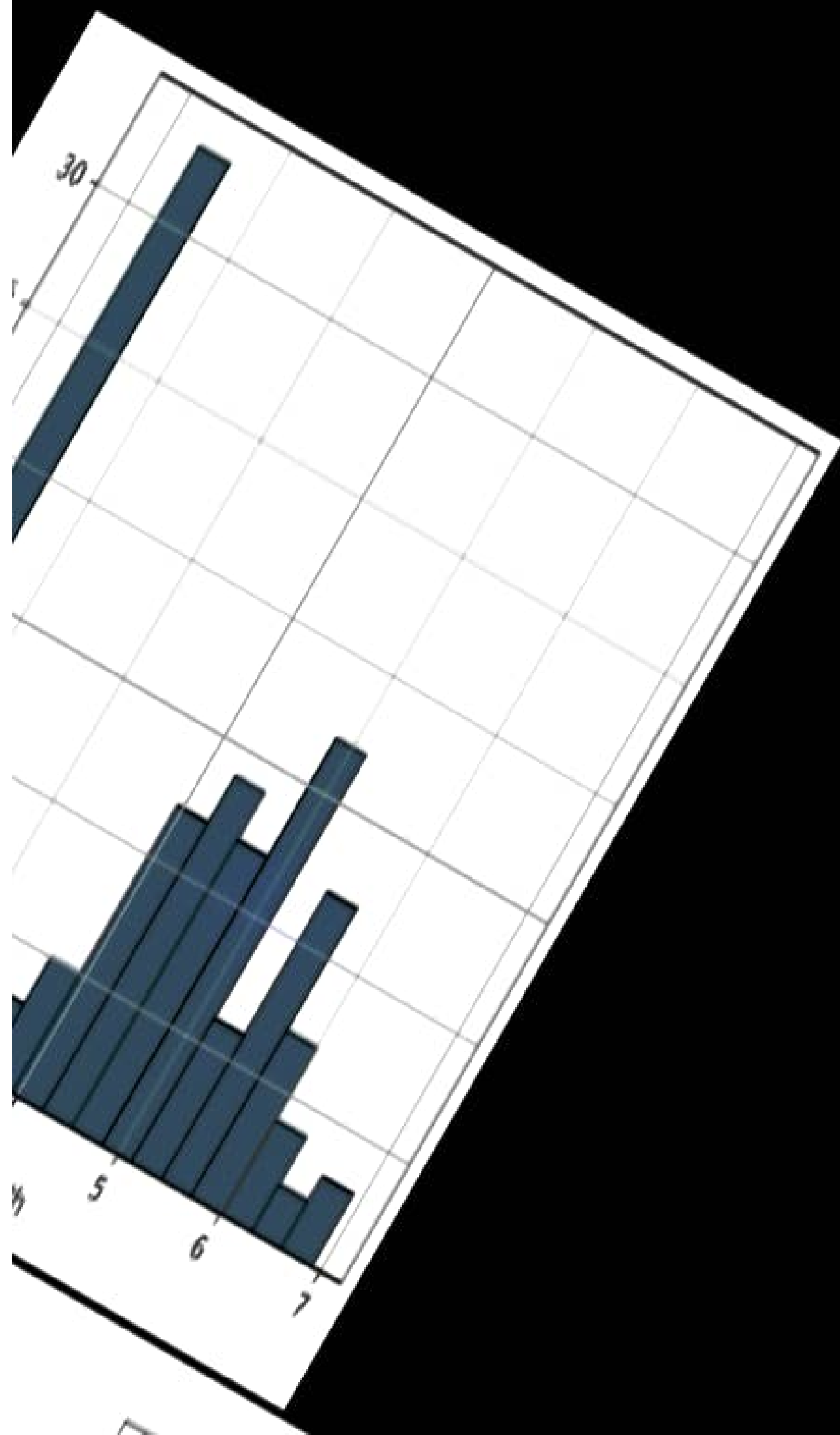


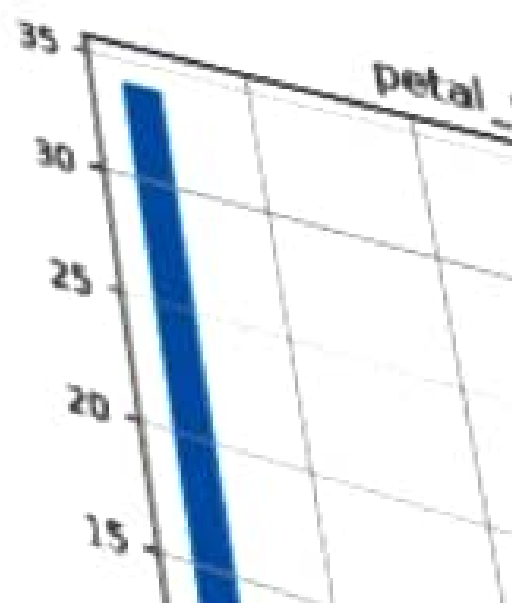
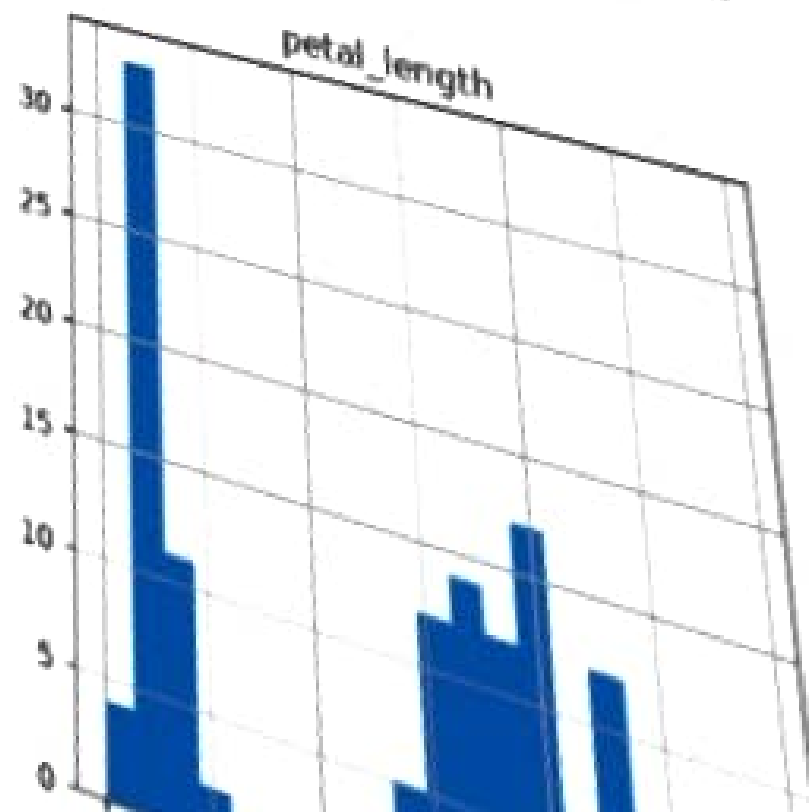
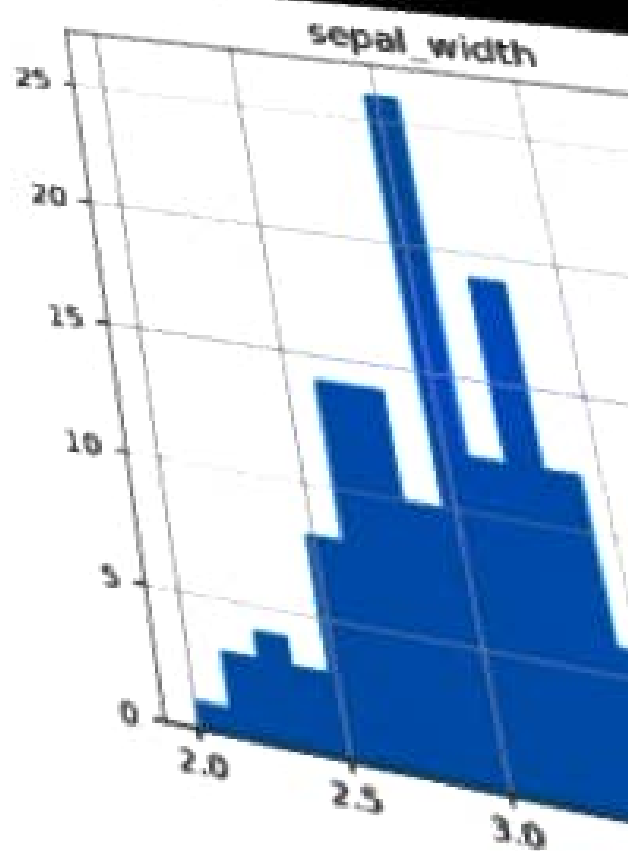
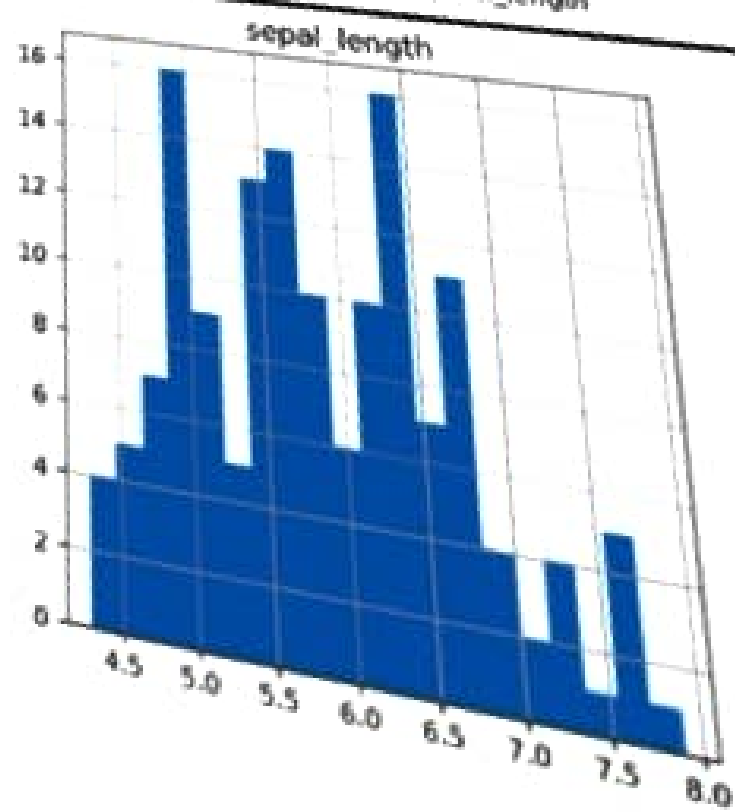
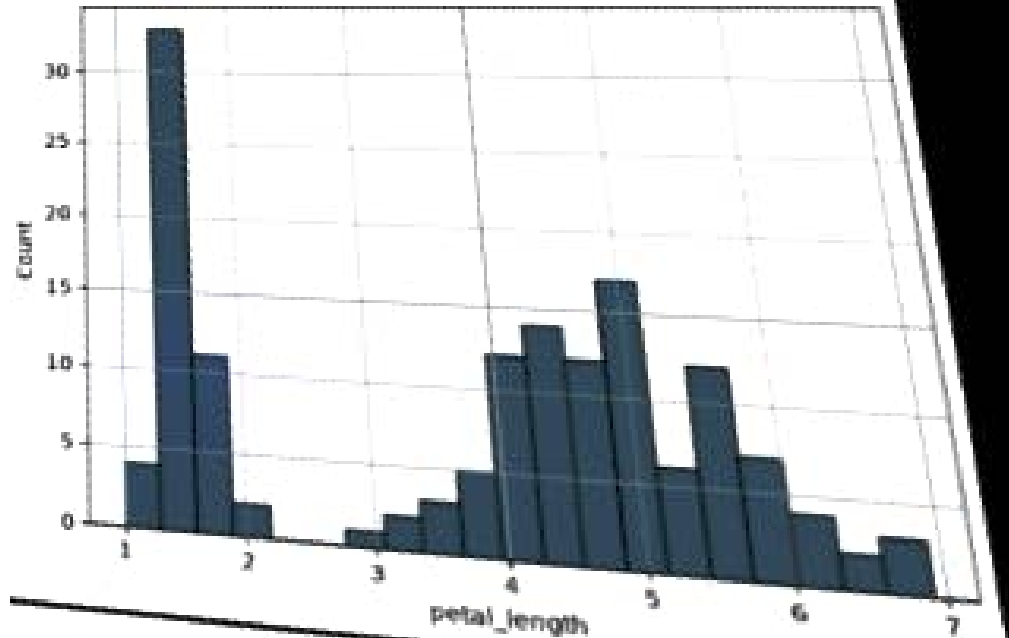
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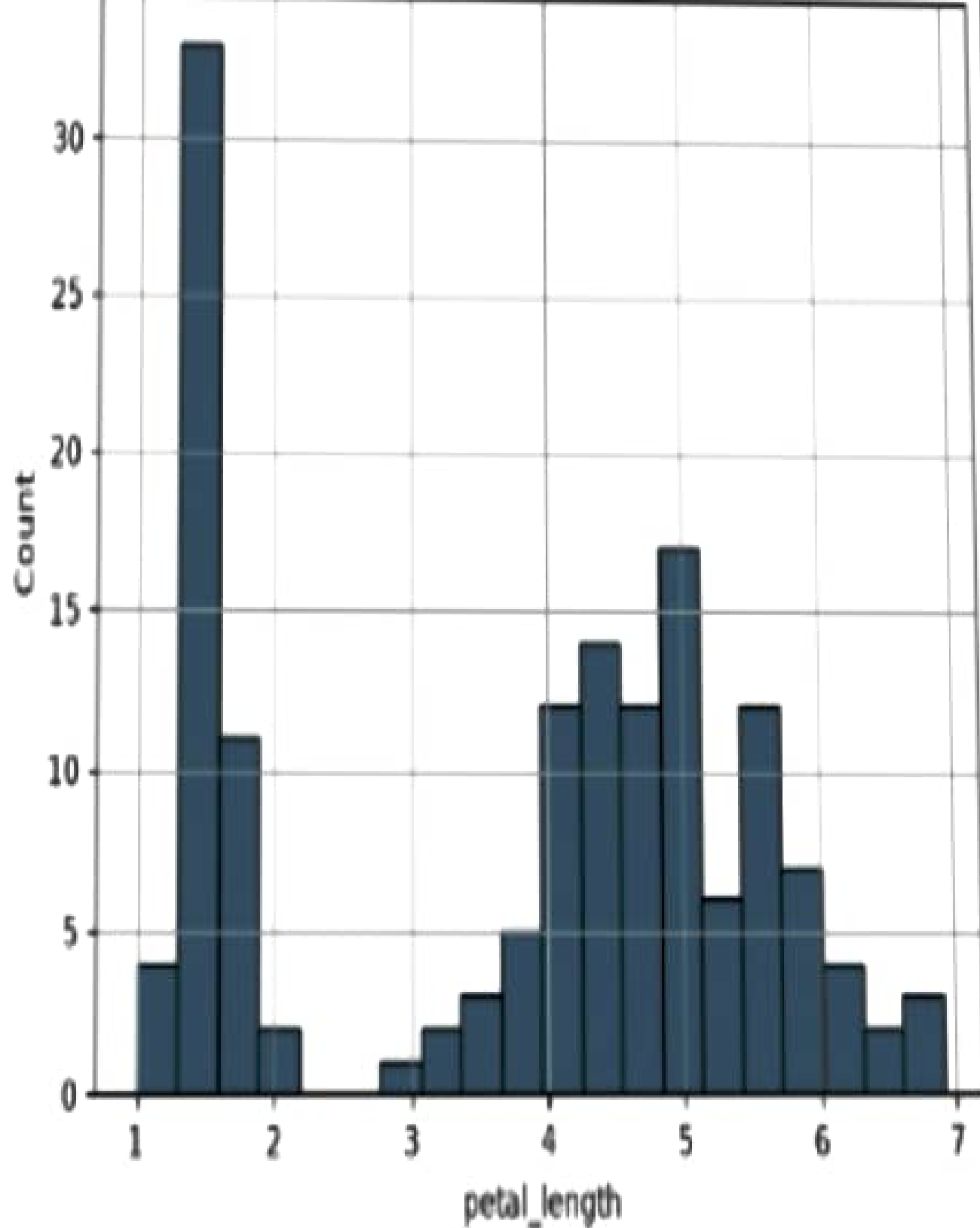


petal_width

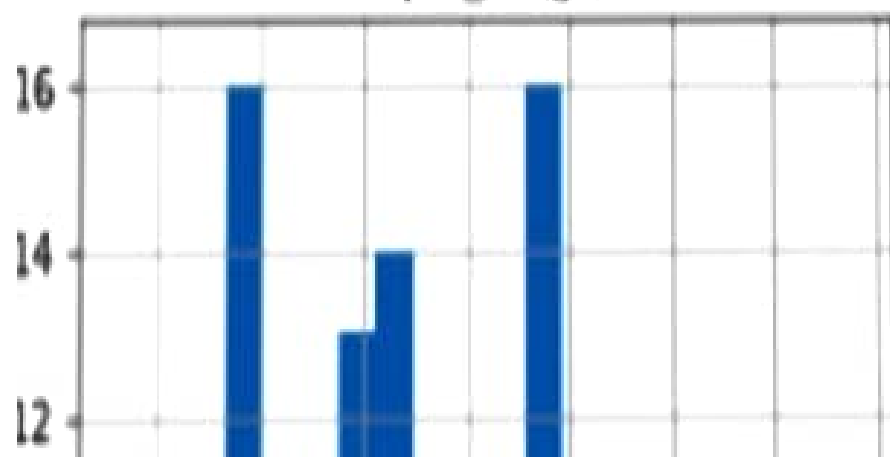






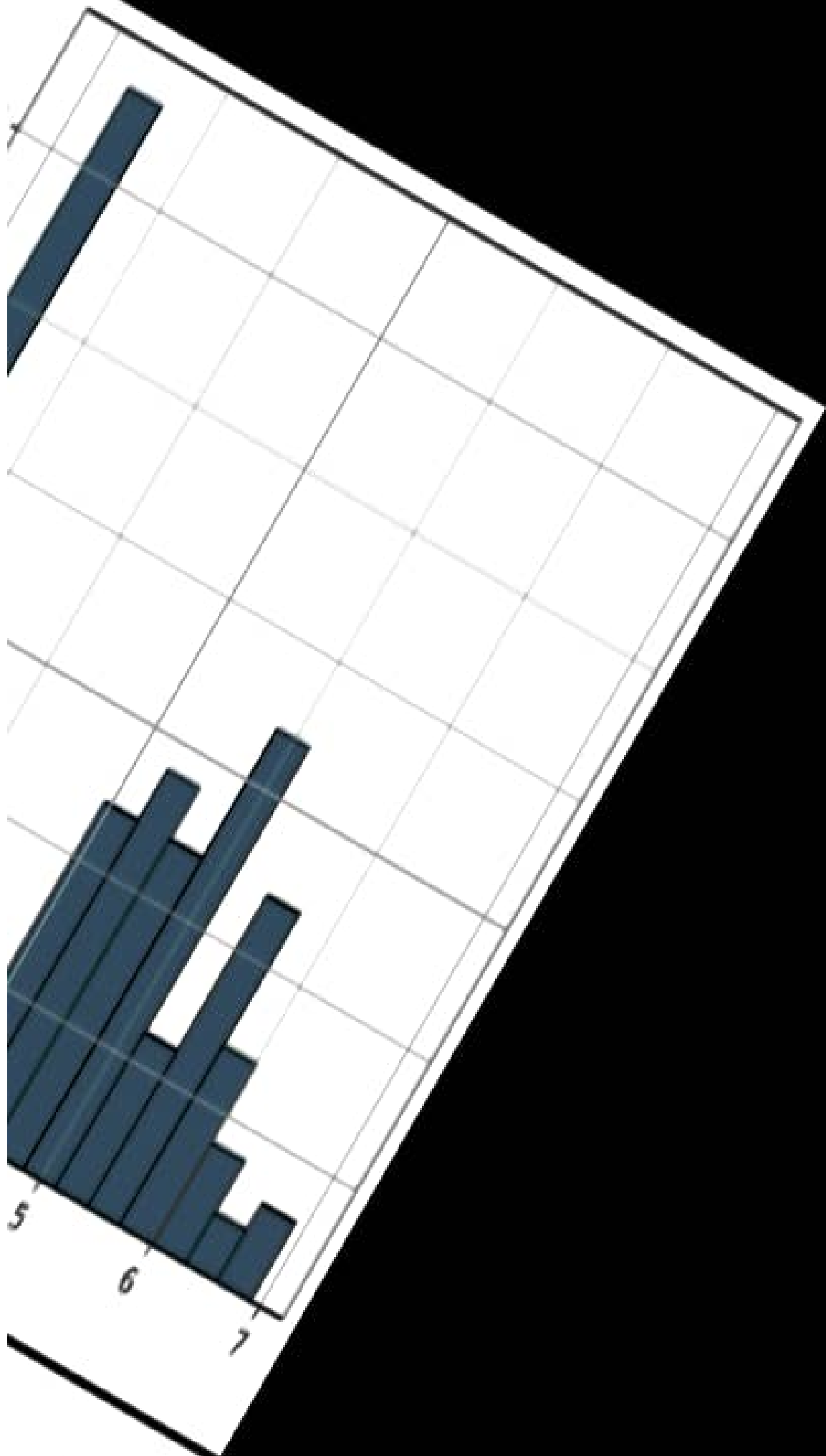


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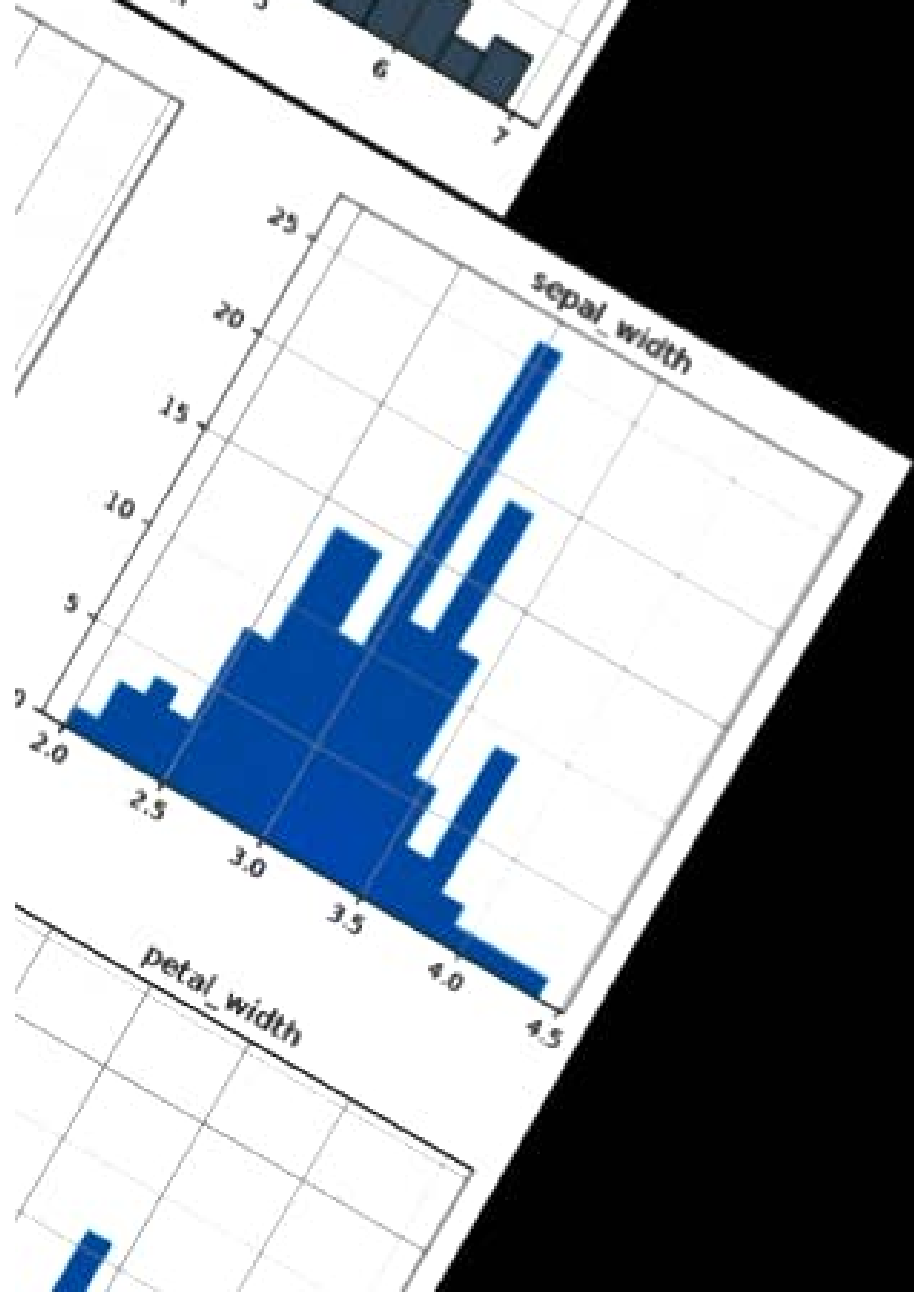
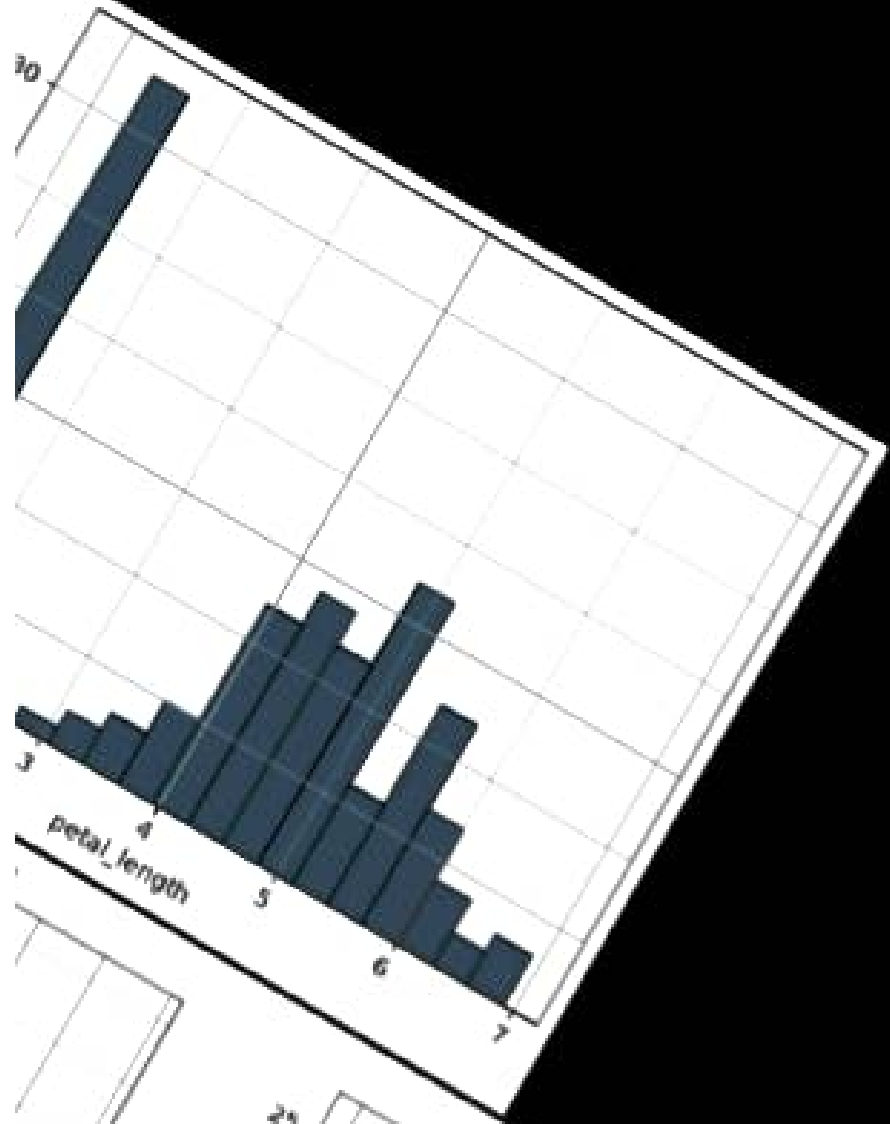


sepal_length





500



Enter translation in X direction: 40
Enter translation in Y direction: 30
Enter rotation angle: 30
Enter X scaling factor: 1.3
Enter Y scaling factor: 2.2
Enter horizontal shear: 0.2
Enter vertical shear: 0.3

Translation Matrix:

$$\begin{bmatrix} 1. & 0. & 40. \\ 0. & 1. & 30. \\ 0. & 0. & 1. \end{bmatrix}$$

Rotation Matrix:

$$\begin{bmatrix} 0.866 & -0.5 & 0. \\ 0.5 & 0.866 & 0. \\ 0. & 0. & 1. \end{bmatrix}$$

Scaling Matrix:

$$\begin{bmatrix} 1.3 & 0. & 0. \\ 0. & 2.2 & 0. \\ 0. & 0. & 1. \end{bmatrix}$$

Shearing Matrix:

$$\begin{bmatrix} 1. & 0.2 & 0. \\ 0.3 & 1. & 0. \\ 0. & 0. & 1. \end{bmatrix}$$

Rotation + Scaling Matrix:

$$\begin{bmatrix} 1.126 & -1.1 & 0. \\ 0.65 & 1.905 & 0. \\ 0. & 0. & 1. \end{bmatrix}$$

Translation + Rotation + Scaling Matrix:

$$\begin{bmatrix} 1.126 & -1.1 & 40. \\ 0.65 & 1.905 & 30. \\ 0. & 0. & 1. \end{bmatrix}$$

All Transformations Matrix:

$$\begin{bmatrix} 0.796 & -0.875 & 40. \\ 1.222 & 2.035 & 30. \\ 0. & 0. & 1. \end{bmatrix}$$